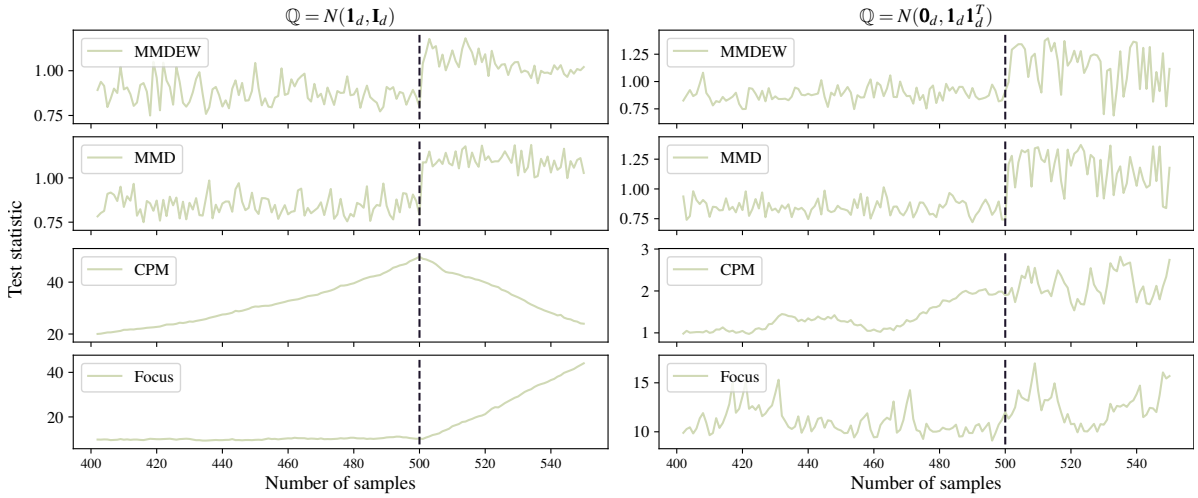

Figure 8: Maximum values of the respective test-statistics (20 repetitions, $d = 20$). A change (indicated by a dashed line) occurs after 500 samples, from $\mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$ to the distribution indicated on top of the columns, respectively. For the univariate approaches (CPM, Focus), we run one instance per dimension and consider the mean.

Our results are in Figure 8. A change in either the distribution mean (l.h.s.) or the correlation (r.h.s.) lead to an increase of the test statistic of MMDEW and MMD, respectively. Hence, these approaches allow detecting such changes. CPM and Focus correctly identify the change in mean, which is reflected in the univariate marginals they consider. CPM correctly identifies the change point, that is, the maximum value of the test statistic is at 500. When regarding the change in the correlation (the marginals the univariate approaches consider do not change), Focus' test statistic does not reflect the change point. Surprisingly, for CPM, the change in the correlation structure leads to a change in the test-statistic—but the change is identified incorrectly, with the maximum of the test statistic occurring after approximately 530 samples. We conclude that using MMD or the proposed MMDEW is preferable to aggregating univariate change detectors when processing multivariate data, when the changes are not reflected in the marginals.